Machine Fact Sheet

Institutional Memory — DAQ Condition Intelligence

Machine identity: the make/model, control system and spindle motor rating are still not stated anywhere in the project's documentation. Everything else on this page is now derived directly from a real header row and 5 real data rows sampled from every one of the 25 condition folders, not guessed — with two corrections to what was assumed on an earlier pass, called out below.

What the three process parameters control

Every recording sits under a condition folder named like `4_S800_F35_D0.25`, which encodes three settings the operator chose before that run:

Symbol	What it is	Plain-language meaning
S	Spindle speed	How fast the cutting tool spins, in RPM. Higher S = faster rotation.
F	Feed rate	How fast the tool moves through the material, in mm/min. Higher F = faster cutting, more load per tooth.
D	Depth of cut	How deep each pass cuts into the material, in mm. Higher D = more material removed per pass, more force.

The experiment isn't random — it's a Latin square

Laying out all 25 conditions by their actual S and F values reveals the depth of cut wasn't chosen arbitrarily: for every spindle speed, the five feed rates (20, 25, 30, 35, 40 mm/min) are paired with the five depths (0.10, 0.15, 0.20, 0.25, 0.30 mm) in a rotating order, so that **every depth appears exactly once at each speed, and exactly once at each feed rate.** That's a textbook 5×5 Latin-square design of experiments — it means no two conditions share more than one setting in common by coincidence, which is exactly what you want when trying to isolate how each parameter affects vibration, load and current independently.

All 25 process conditions, with real recording counts

Counted directly from the dataset on 2025-09-30. The counts total exactly **822**, matching the corpus size already reported in README.md §1.

Cond.	Folder	S (RPM)	F (mm/min)	D (mm)	Recs.
1	1_S800_F20_D0.10	800	20	0.10	35
2	2_S800_F25_D0.15	800	25	0.15	31
3	3_S800_F30_D0.20	800	30	0.20	36
4	4_S800_F35_D0.25	800	35	0.25	39
5	5_S800_F40_D0.30	800	40	0.30	35
6	6_S900_F20_D0.15	900	20	0.15	30
7	7_S900_F25_D0.20	900	25	0.20	32
8	8_S900_F30_D0.25	900	30	0.25	35
9	9_S900_F35_D0.30	900	35	0.30	33

Cond.	Folder	S (RPM)	F (mm/min)	D (mm)	Recs.
10	10_S900_F40_D0.1	900	40	0.10	36
11	11_S1000_F20_D0.20	1000	20	0.20	24
12	12_S1000_F25_D0.25	1000	25	0.25	25
13	13_S1000_F30_D0.30	1000	30	0.30	24
14	14_S1000_F35_D0.10	1000	35	0.10	34
15	15_S1000_F40_D0.15	1000	40	0.15	35
16	16_S1100_F20_D0.25	1100	20	0.25	31
17	17_S1100_F25_D0.30	1100	25	0.30	34
18	18_S1100_F30_D0.1	1100	30	0.10	39
19	19_S1100_F35_D0.15	1100	35	0.15	35
20	20_S1100_F40_D0.20	1100	40	0.20	34
21	21_S1200_F20_D0.30	1200	20	0.30	28
22	22_S1200_F25_D0.10	1200	25	0.10	32
23	23_S1200_F30_D0.15	1200	30	0.15	35
24	24_S1200_F35_D0.20	1200	35	0.20	34
25	25_S1200_F40_D0.25	1200	40	0.25	36
				TOTAL	822

Recording counts range from 24 (conditions 11 and 13) to 39 (conditions 4 and 18) — not a flat count per condition as assumed on an earlier pass.

The six sensor channels — correction: these are raw volts, not physical units

Every CSV has a second row, right after the column-name header, that reads `Unit,V,V,V,V,V,V`. That's not sensor data — it's a one-line legend, and it says plainly that all six channels are recorded as **raw voltage output (V)**, not converted engineering units. An earlier pass at this fact sheet assumed mm/s², N and A; that was a guess and it was wrong. Unless a calibration/scale factor is documented elsewhere, treat every reading as a raw transducer voltage, not a calibrated physical quantity.

Channel	What it measures	Unit in file	Typical sensor placement (inferred — confirm)
Vib_Spindle	Vibration at the spindle	V (raw)	Accelerometer mounted on the spindle housing/head
Vib_Table	Vibration at the machine table	V (raw)	Accelerometer mounted on the table or workpiece fixture
X_Load_Cell	Cutting force along the X axis	V (raw)	Force/dynamometer sensor under the workpiece fixture
Y_Load_Cell	Cutting force along the Y axis	V (raw)	Same fixture, Y-axis channel
Z_Load_Cell	Cutting force along the Z axis	V (raw)	Same fixture, Z-axis channel
Current	Electrical current drawn by the spindle motor	V (raw)	Current clamp/sensor on the spindle drive line, output as a proportional voltage

Sensor placement descriptions are still typical setups for this kind of instrumentation, inferred from the channel names — a voltage output alone doesn't tell you where the sensor is physically mounted. Replace with the actual mounting/calibration documentation if available.

Sampling rate — observed, not yet fully verified

In every one of the 25 sampled files, the Time column steps by exactly 0.001 between rows (0, 0.001, 0.002, 0.003 …), consistent across every condition checked. If Time is in seconds, that's a **~1000 Hz sampling rate**. This is based on the first 5 rows of one recording per condition, not every file — the pipeline's own design already measures the rate per-recording from real Time deltas rather than assuming a fixed value (README.md §2), which is the right approach; treat 1000 Hz as a strong prior to sanity-check against, not a hardcoded constant.

Source: header row + 5 sample data rows read directly from Recording 1.csv in each of the 25 condition folders, 2025-09-30, plus README.md §1–§2. Machine model and exact sensor placement/calibration remain undocumented and should be filled in.